Department of
Computer Science and Engineering

Title: Implementation of Interface and Multiple Inheritance

Object Oriented Programming Lab

CSE 202

Green University of Bangladesh

1 Objective(s)

- To gather knowledge of how interface works.
- Understand multiple inheritance using a class implementing multiple interfaces.
- To solve problems using the concept of interface.

2 Problem analysis

An interface is a reference type in Java. It is similar to class. But it is a collection of abstract methods and static constants. Three important attributes of interfaces are :

1. Access of an interface is either public or default.
2. Variables declared inside an interface are implicitly final and static.
3. All methods and variables are implicitly public if the interface, itself, is declared as public.

Interfaces provide specifications that a class (which implements it) must follow. Interfaces are also used to achieve multiple inheritance in Java. If a subclass is inherited from two or more classes, it's multiple inheritance. Using the concept of interface, implement the following:

- Creating an interface Printable with an abstract method - void print();
- Creating another interface Showable with an abstract method - void show();
- Create a Drawable class which implements both Printable and Showable interfaces and implements their methods.
- Create an object of Drawable class in the main method and call the print and show methods.

3 Keywords

3.1 interface

Like the **class** keyword, the **interface** keyword is used to declare an interface.

```java
interface Polygon{
    public static final String color = "Green";
    public void getArea();
}
```

In the example above, we have used the **interface** keyword to declare an interface named Polygon. It includes a constant variable color and an abstract method getArea(). This means that any class that implements *Polygon* must provide an implementation for the **getArea()** method.

3.2 implements

Like abstract classes, we cannot create objects of interfaces. However, we can implement interfaces in other classes. In Java, we use the **implements** keyword to implement interfaces. To implement an interface a class must define each of the method declared in the interface. Each class can also add new features.

```java
interface Polygon{
    public static final String color = "Green";
    public void getArea();
}

class Rectangle implements Polygon{
    public void getArea(int length, int breadth){
        System.out.println("The area of the rectangle is "+ (length *
            breadth));
    }
}
```

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In the above program, the *Rectangle* class implements *Polygon* interface, therefore it has the method **getArea()** with implementation.

3.3 extends

Similar to classes, interfaces can extend other interfaces. The extends keyword is used for extending interfaces.

```
interface Line{
    //members of Line interface
}

interface Polygon extends Line{
    //members of Polygon interface and Line interface
}
```

As shown in the figure given below, a class extends another class, an interface extends another interface but a class implements an interface.

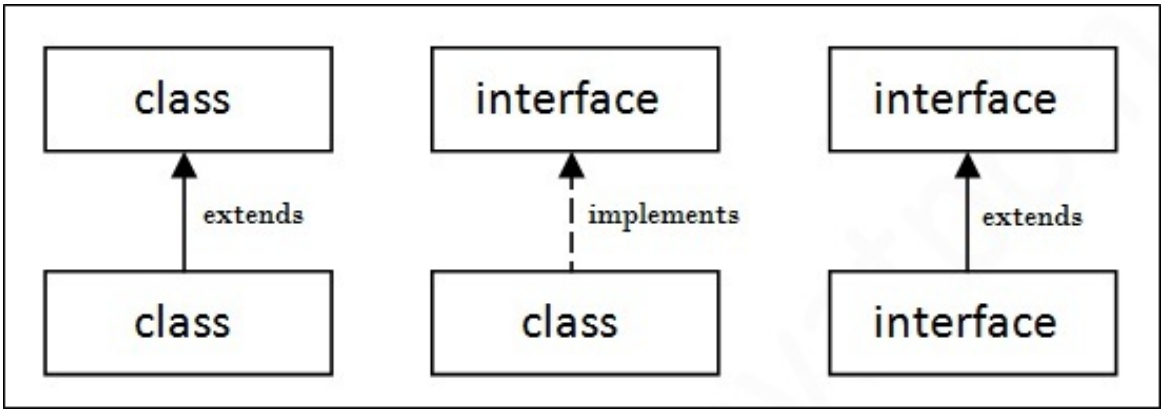

Figure 1: Relationship between inteface and class

3.4 Multiple Inheritance

If a class implements multiple interfaces, or an interface extends multiple interfaces i.e. known as multiple inheritance.

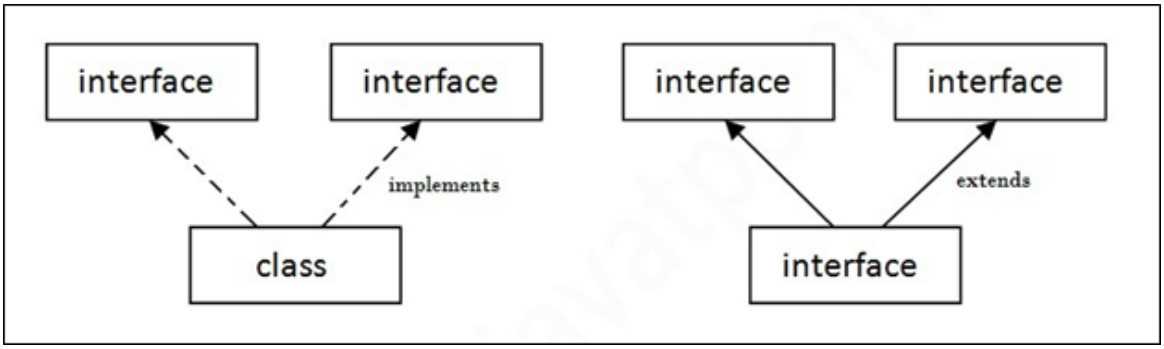

Figure 2: Multiple inheritance using Interface

4 Implementation in Java

```
interface Printable{
    void print();
```

```
}

interface Showable{
    void show();
}

class Drawable implements Printable,Showable{

public void print(){System.out.println("Hello");}
public void show(){System.out.println("Welcome");}
}

class Main{
    public static void main(String args[]){
        Drawable obj = new Drawable();
        obj.print();
        obj.show();
    }
}
```

5 Input/Output

Output of the program is given below.

```
Hello
Welcome
```

6 Discussion & Conclusion

The problem demonstrates multiple inheritance using a class implementing multiple interfaces.

7 Lab Task (Please implement yourself and show the output to the instructor)

1. Implement Multiple Inheritance Using An Interface Extending Multiple Interfaces.

7.1 Problem analysis

- Create two interfaces A and B.
- Create an interface C which extends interface A and B
- Create a class D which implements C.
- Create an object of class D in the main method and call all available methods from the interfaces.

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8 Lab Exercise (Submit as a report)

- Create an interface isEmergency with only one method - soundSiren which takes no arguments and returns no value.
- Write a class FireEmergency that implements the IsEmergency interface. The soundSiren method should print "Siren Sounded".
- Write a class SmokeAlarm that does not implement any interface. The class has an empty body.
- Create an array of Object class, myArray in the main method.
- Construct 2 SmokeAlarm object and add it to the array myArray in the main method.
- Construct 2 FireEmergency object and add it to the array myArray in the main method.
- In the main method, write a for loop, to print which array elements are instances of classes that implement the IsEmergency interface and if so, call the soundSiren method.

9 Policy

Copying from internet, classmate, seniors, or from any other source is strongly prohibited. 100% marks will be *deducted* if any such copying is detected.

10 References

https://www.programiz.com/java-programming/interfaces

11 Appendix

If required, course instructor might provide some additional materials.

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